

MATH 501: REAL ANALYSIS

RUIZHE QIAN

CONTENTS

1. August 25, 2026	1
1.1. Measures	1
1.2. Algebras and σ -algebras	3
1.3. Borel σ -algebras	4

1. AUGUST 25, 2026

Real analysis will be developed through measure theory, Lebesgue integration, and the study of function spaces such as L^p spaces.

1.1. Measures. The goal of a measure is to assign a length, area, or volume to subsets of $\mathbb{R}$, $\mathbb{R}^2$, or, more generally, $\mathbb{R}^n$. Ideally, we would like a function

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

with the following properties.

- (i) *Countable additivity:* if $E_1, E_2, \dots \subseteq \mathbb{R}^n$ are pairwise disjoint, then

$$m\left(\bigcup_{j=1}^{\infty} E_j\right) = \sum_{j=1}^{\infty} m(E_j).$$

A weaker requirement is finite additivity, in which the identity is required only for finite pairwise disjoint families.

- (ii) *Invariance under rigid motions:* if

$$\tilde{E} = AE + b, \quad A \in O(n), \quad b \in \mathbb{R}^n,$$

then $m(\tilde{E}) = m(E)$. Here A represents a rotation or reflection, and b a translation.

(iii) *Normalization:*

$$m([0, 1]^n) = 1.$$

Unfortunately, a measure with all these properties cannot be defined on every subset of $\mathbb{R}^n$.

Theorem 1 (Vitali). *Assuming the axiom of choice, there is no countably additive, rigid-motion-invariant map*

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

satisfying $m([0, 1]^n) = 1$.

Proof. It is enough to consider $n = 1$: if such a map existed on $\mathbb{R}^n$, then

$$\mu(E) = m(E \times [0, 1]^{n-1}), \quad E \subseteq \mathbb{R},$$

would have the required properties on $\mathbb{R}$. We therefore write m for this one-dimensional map.

First, $m(\{x\}) = 0$ for every $x \in \mathbb{R}$. Otherwise, translation invariance and countable additivity applied to a countably infinite set in $[0, 1]$ would contradict $m([0, 1]) = 1$. Consequently, $m([0, 1)) = 1$.

Define an equivalence relation on $[0, 1)$ by

$$x \sim y \iff x - y \in \mathbb{Q}.$$

Using the axiom of choice, select one representative from each equivalence class and let V be the set of representatives. Thus, for every $x \in [0, 1)$, there is a unique $y \in V$ such that $x - y \in \mathbb{Q}$.

For $t \in [0, 1)$, define the translate of V modulo 1 by

$$\begin{aligned} V_t &= V + t \pmod{1} \\ &= \{x + t : x \in V, x + t < 1\} \\ &\quad \cup \{x + t - 1 : x \in V, x + t \geq 1\}. \end{aligned}$$

Finite additivity and translation invariance give

$$m(V_t) = m(V).$$

The family indexed by $t \in [0, 1) \cap \mathbb{Q}$ is a partition of $[0, 1)$. Indeed, if $x \in [0, 1)$ and $y \in V$ represents the equivalence class of x , then either

$t = x - y$ or $t = x - y + 1$ lies in $[0, 1) \cap \mathbb{Q}$ and $x \in V_t$. On the other hand, if $V_t \cap V_{t'} \neq \emptyset$, the uniqueness of representatives implies $t = t'$.

Countable additivity now yields

$$1 = m([0, 1)) = \sum_{t \in [0, 1) \cap \mathbb{Q}} m(V_t) = \sum_{t \in [0, 1) \cap \mathbb{Q}} m(V).$$

If $m(V) = 0$, the last sum is 0; if $m(V) > 0$, it is infinite. Either possibility is a contradiction. $\square$

One might try to rescue the construction by requiring only finite additivity together with invariance and normalization. In dimensions at least three, even this is impossible.

Theorem 2 (Banach–Tarski). *If $n \geq 3$, there is no finitely additive, rigid-motion-invariant map*

$$m: 2^{\mathbb{R}^n} \longrightarrow [0, \infty]$$

satisfying $m([0, 1]^n) = 1$.

The resolution is to give up the requirement that a measure be defined on *all* subsets of $\mathbb{R}^n$. In fact, measure theory can be developed on a general underlying set X once an appropriate collection of measurable subsets has been chosen.

1.2. Algebras and σ -algebras.

Definition 1 (Algebra). *An algebra on a set X is a nonempty collection $\mathcal{A} \subseteq 2^X$ such that*

- (a) *if $E_1, \dots, E_N \in \mathcal{A}$, then $\bigcup_{j=1}^N E_j \in \mathcal{A}$;*
- (b) *if $E \in \mathcal{A}$, then $E^c = X \setminus E \in \mathcal{A}$.*

Definition 2 (σ -algebra). *A σ -algebra on a set X is a nonempty collection $\mathcal{A} \subseteq 2^X$ such that*

- (a) *if $E_n \in \mathcal{A}$ for every $n \in \mathbb{N}$, then $\bigcup_{n=1}^{\infty} E_n \in \mathcal{A}$;*
- (b) *if $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.*

Because the collections in these definitions are nonempty and closed under complements, they contain both X and $\emptyset$. By De Morgan's

laws, an algebra is closed under finite intersections and a σ -algebra is closed under countable intersections. For example,

$$\bigcap_{n=1}^{\infty} E_n = \left(\bigcup_{n=1}^{\infty} E_n^c \right)^c.$$

When an algebra is already known to be closed under finite set operations, it is enough to check closure under countable *disjoint* unions in order to prove that it is a σ -algebra. Given $E_n \in \mathcal{A}$, set

$$\tilde{E}_1 = E_1, \quad \tilde{E}_n = E_n \setminus \bigcup_{j=1}^{n-1} E_j \quad (n \geq 2).$$

Then the sets $\tilde{E}_n$ are pairwise disjoint and

$$\bigcup_{n=1}^{\infty} \tilde{E}_n = \bigcup_{n=1}^{\infty} E_n.$$

Example 1. The collection $\{\emptyset, X\}$ is the smallest σ -algebra on X .

Example 2. The power set 2^X is the largest σ -algebra on X .

Example 3 (Countable-cocountable σ -algebra). The collection

$$\mathcal{A} = \{E \subseteq X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ -algebra on X .

A convenient way to specify a σ -algebra is to give a family of generators.

Definition 3 (Generated σ -algebra). Let $\mathcal{E} \subseteq 2^X$. The σ -algebra generated by $\mathcal{E}$, denoted $\mathcal{M}(\mathcal{E})$, is the smallest σ -algebra containing $\mathcal{E}$.

This object exists because 2^X is a σ -algebra containing $\mathcal{E}$, and an arbitrary intersection of σ -algebras is again a σ -algebra. Explicitly,

$$\mathcal{M}(\mathcal{E}) = \bigcap \{\mathcal{A} : \mathcal{A} \text{ is a } \sigma\text{-algebra on } X \text{ and } \mathcal{E} \subseteq \mathcal{A}\}.$$

1.3. Borel σ -algebras.

Definition 4 (Borel σ -algebra). Let X be a metric space, or more generally a topological space. The Borel σ -algebra $\mathcal{B}_X$ is the σ -algebra generated by the open subsets of X .

For example, every open subset of $\mathbb{R}$ is a union of open intervals. The Borel σ -algebra contains every open and closed set, every G_δ set (a countable intersection of open sets), every F_σ set (a countable union of closed sets), and all sets obtained by continuing such countable operations. These classes form the Borel hierarchy.

Proposition 1. *The Borel σ -algebra $\mathcal{B}_\mathbb{R}$ is generated by any one of the following collections:*

- (a) $\{(a, b) : a < b\};$
- (b) $\{[a, b] : a < b\};$
- (c) $\{(-\infty, a] : a \in \mathbb{R}\}.$

Proof. Let the three collections above be $\mathcal{E}_1$, $\mathcal{E}_2$, and $\mathcal{E}_3$, respectively.

Every member of $\mathcal{E}_1$ is open, so $\mathcal{M}(\mathcal{E}_1) \subseteq \mathcal{B}_\mathbb{R}$. For the reverse inclusion, let $U \subseteq \mathbb{R}$ be open. For every $x \in U$, there is an $\varepsilon > 0$ such that

$$(x - \varepsilon, x + \varepsilon) \subseteq U.$$

By the density of $\mathbb{Q}$ in $\mathbb{R}$, we can choose $p, q \in \mathbb{Q}$ such that

$$x - \varepsilon < p < x < q < x + \varepsilon.$$

Thus every $x \in U$ lies in an interval (p, q) with rational endpoints that is contained in U . It follows that

$$U = \bigcup \{(p, q) : p, q \in \mathbb{Q}, p < q, (p, q) \subseteq U\}.$$

The indexing collection is a subset of $\mathbb{Q}^2$, so it is countable. Therefore every open subset of $\mathbb{R}$ is a countable union of open intervals. In particular, every open set belongs to $\mathcal{M}(\mathcal{E}_1)$, proving

$$\mathcal{M}(\mathcal{E}_1) = \mathcal{B}_\mathbb{R}.$$

Every closed interval is Borel. In the other direction, for $a < b$,

$$(a, b) = \bigcup_{\substack{n \in \mathbb{N} \\ 2/n < b-a}} \left[a + \frac{1}{n}, b - \frac{1}{n} \right].$$

Thus $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_2)$, and consequently $\mathcal{M}(\mathcal{E}_2) = \mathcal{B}_{\mathbb{R}}$. Equivalently, the reverse containment can also be seen from

$$[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n} \right).$$

Finally, every left half-line in $\mathcal{E}_3$ is closed and hence Borel. Conversely,

$$(-\infty, b) = \bigcup_{n=1}^{\infty} \left(-\infty, b - \frac{1}{n} \right]$$

and therefore

$$(a, b) = (-\infty, b) \cap (-\infty, a]^c \in \mathcal{M}(\mathcal{E}_3).$$

It follows that $\mathcal{E}_1 \subseteq \mathcal{M}(\mathcal{E}_3)$, so $\mathcal{M}(\mathcal{E}_3) = \mathcal{B}_{\mathbb{R}}$ as well. $\square$